PROJECT REPORT

1. Introduction & Problem Statement

In modern retail, e-commerce, and supply-chain environments, accurate sales forecasting is essential for effective inventory planning, demand–supply optimization, and revenue estimation. Businesses that fail to predict demand accurately often face challenges such as stockouts, excess inventory, and increased operational costs.

Traditional regression models treat data points as independent observations and often ignore the temporal nature of sales data. However, sales data is inherently time-dependent and is influenced by trends, seasonality, and past demand patterns.

The objective of this project is to design and implement a time-aware machine learning system that forecasts future sales using regression techniques combined with time-based feature engineering. The system also evaluates forecasting performance using appropriate metrics and converts predictions into actionable business insights.

2. Dataset Description & Data Understanding

The project uses historical sales data stored in CSV format. The dataset contains:

- A **date/time column** representing when the sale occurred
- One or more **numerical columns**, including the target variable (sales)
- Optional additional numeric predictors

Data Understanding Steps:

- The dataset is loaded dynamically using a user interface.
- The date column is converted into datetime format and sorted chronologically.
- Time granularity (daily data) is inferred from the date column.
- Summary statistics such as mean, minimum, and maximum values are examined.
- Sales trends are visualized to identify seasonality, trends, and volatility.

This step ensures the data is suitable for time-series forecasting and helps identify anomalies or missing patterns before modeling.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand sales behavior over time.

Key analyses include:

- Visualization of historical sales trends
- Identification of short-term fluctuations and long-term trends
- Observation of recurring seasonal patterns such as weekly or monthly demand cycles

These insights guide the feature engineering process and help in selecting suitable modeling techniques.

4. Feature Engineering (Core Component)

Feature engineering is the most critical part of this project, as it enables regression models to learn temporal dependencies.

4.1 Date-Based Features

From the date column, the following features are extracted:

- **Day of Week** – captures weekly seasonality
- **Month** – captures monthly and seasonal demand patterns

4.2 Lag Features

Lag features allow the model to learn from past sales values:

- **Lag 1**: Sales from the previous day
- **Lag 7**: Sales from the previous week
- **Lag 30**: Sales from the previous month

These features introduce memory into the regression model.

4.3 Rolling Statistical Features

Rolling statistics capture recent trends and volatility:

- 7-day rolling mean and standard deviation
- 30-day rolling mean and standard deviation

These features help the model understand short-term and medium-term demand behavior.

4.4 Avoiding Feature Leakage

- All lag and rolling features are created using only past data.

- Rows with insufficient historical data are removed.
- Future data is never used during training.

This ensures the model remains realistic and production-ready.

5. Model Development

Two regression models are implemented and compared:

5.1 Linear Regression (Baseline Model)

- Serves as a simple baseline.
- Helps evaluate how much performance improves when time-aware features are added.

5.2 Random Forest Regressor (Enhanced Model)

- Captures non-linear relationships.
- Performs well with engineered lag and rolling features.
- Supports feature importance analysis.

Model Selection Justification:

Random Forest is chosen as the primary model due to its ability to handle non-linear interactions between time-based features and sales values.

6. Time-Aware Train–Test Split

Instead of random splitting, a chronological split is used:

- First 80% of data → Training set
- Remaining 20% → Testing set

This simulates real-world forecasting where future sales must be predicted using only past information.

7. Model Evaluation

Models are evaluated using forecasting-specific metrics:

- **Mean Absolute Error (MAE)** – average absolute prediction error
- **Root Mean Squared Error (RMSE)** – penalizes larger errors
- **Mean Absolute Percentage Error (MAPE)** – relative percentage error

The enhanced model with time-based features consistently outperforms the baseline model, demonstrating the importance of temporal feature engineering.

8. Forecasting Future Sales

Using the best-performing model, future sales are forecasted for a user-defined number of days.

Recursive Forecasting Approach:

- Each predicted value is fed back into the model as a lag input.
- Lag and rolling features are updated dynamically.
- This closely mimics real-world forecasting scenarios.

The forecasted results are visualized alongside recent historical data for easy interpretation.

9. Demand Optimization & Business Insights

Based on the forecast:

- High-demand periods can be identified in advance.
- Inventory levels can be increased during expected demand peaks.
- Overstocking risk can be reduced during low-demand periods.
- Anomaly detection highlights unusual spikes or drops in sales.

These insights help businesses make informed operational and inventory decisions.

10. Bonus Features (Value Addition)

The system also includes:

- **Anomaly Detection** using Isolation Forest, LOF, and Z-Score
- **Explainable AI (XAI)** using feature importance and SHAP values
- **Interactive GUI Dashboard** for non-technical users

These components enhance real-world applicability but are not required for core evaluation.

11. Conclusion

This project successfully demonstrates how time-aware feature engineering significantly improves sales forecasting accuracy using regression models. By combining lag features, rolling statistics, and proper evaluation techniques, the system produces reliable forecasts and actionable business insights. The project adheres strictly to time-series modeling principles and avoids common pitfalls such as data leakage and random train-test splitting.